

LOW ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report evaluates the patent evidence provided for low acrylonitrile nitrile butadiene rubber (NBR) synthesis and related elastomeric material disclosures. The target definition is a butadiene-acrylonitrile copolymer with reduced acrylonitrile content, typically below about 20 wt% ACN, while retaining elastomeric performance. In the supplied manifest, only one primary patent was identified: US10443339B2, which is not a composition or polymerization patent for NBR itself, but rather a downhole elastomeric materials patent focused on carbon-based nanoplatelet-reinforced elastomeric components for high-temperature/high-pressure oilfield service. The patent explicitly references NBR among several candidate elastomers used in packer and seal applications, but it does not disclose NBR polymerization, monomer feed ratios, acrylonitrile content ranges, or synthesis conditions for low-ACN NBR.

Accordingly, the patent landscape in this dataset does not provide direct synthetic guidance for low-acrylonitrile NBR manufacture. Instead, it highlights the application context where NBR may be used as a base elastomer in harsh environments and where reinforcement with carbon-based nanoplatelets is proposed to improve performance. The disclosed formulation trend is toward elastomer matrix reinforcement rather than copolymer composition tuning. The process trend is component fabrication by mixing elastomer(s) with nanoscale reinforcement, followed by curing/cross-linking and installation in bore tools or packer assemblies.

From a low-ACN NBR synthesis perspective, the evidence is therefore indirect: it confirms the relevance of NBR in demanding sealing applications, but it does not support specific polymerization recommendations. No emulsion or solution polymerization parameters, initiator systems, emulsifiers, chain-transfer agents, temperatures, pressures, or conversion targets are disclosed for NBR preparation in the provided patent text.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10443339B2
- Patent Title: Elastomeric materials
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose synthesis of low-acrylonitrile NBR. It describes elastomeric components for bore tools and packer assemblies in which an elastomeric matrix is reinforced with carbon-based nanoplatelets; the elastomer may include rubber materials such as NBR, HNBR, AFLAS, or EPDM. The method emphasis is on forming, fitting, and deploying reinforced elastomeric components rather than polymerizing the NBR itself. - Carbon-based nanoplatelets content: less than approximately 10 percent by weight — Claim 1 - Young's modulus: about 3 MPa — Detailed description (ambient rubber example) - HNBR mechanical property decrease temperature: about 160 degrees C. — Detailed description - AFLAS mechanical property decrease temperature: about 220 degrees C. — Detailed description - HPHT environment pressure: up to about 138 MPa — Detailed description - HPHT environment temperature: up to about 205 degrees C. — Detailed description - Ultra-HPHT environment pressure: up to about 241 MPa — Detailed description - Ultra-HPHT environment temperature: up to about 260 degrees C. — Detailed description - HPHT-hc environment pressure: greater than about 241 MPa — Detailed description - HPHT-hc environment temperature: greater than about 260 degrees C. — Detailed description

Relevant Experimental Evidence - Example 1: A bore tool with an elastomeric component containing carbon-based nanoplatelets is disclosed for downhole service. - Example 2: The patent teaches forming an elastomeric component by mixing elastomer(s) with nanoscale reinforcement and curing/cross-linking. - Example 3: Packer assemblies are described with elastomeric elements that may include NBR or other rubbers for harsh environments. - Example 4: The disclosure emphasizes improved performance in high-temperature and chemically harsh bore environments.

Technical Relevance This patent is relevant only as an application-context reference because it identifies NBR as a candidate elastomer for harsh-service seals and packers. It does not advance low-acrylonitrile NBR synthesis directly, but it suggests the performance demands that low-ACN NBR formulations may need to meet.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The provided evidence does not contain a direct low-acrylonitrile NBR synthesis patent. US10443339B2 is best interpreted as a downstream application patent for reinforced elastomeric components, where NBR is mentioned as one possible elastomer but no NBR polymerization or composition-control data are disclosed. For low-ACN NBR synthesis, the evidence base here is insufficient to recommend feed ratios or polymerization conditions; however, it supports the need for formulations that preserve elastomeric integrity in HPHT and chemically aggressive environments. Based on the available evidence, no synthesis parameters can be recommended beyond the general requirement that any low-ACN NBR intended for such service should be compatible with reinforcement and curing strategies used in downhole elastomer components.

5. REFERENCES

- US10443339B2 | Elastomeric materials | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10443339B2/en>